

Eigenvalues and Eigenvectors

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1 Eigenvalues and Eigenvectors



A transformation $\mathbf{x} \mapsto A\mathbf{x}$ may move vectors in a variety of directions. Let $A = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}$, $\mathbf{u} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$. Special vectors are transformed by A into scalar multiples of themselves, e.g. $A\mathbf{v} = 2\mathbf{v}$.

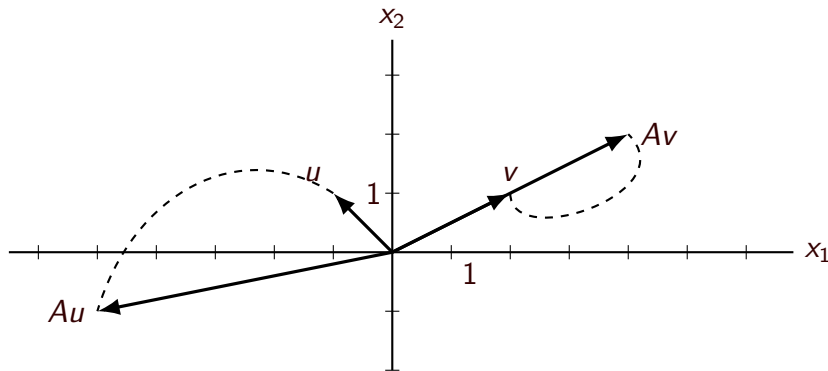


Figure 1. Effects of multiplication by A .

Definition

An **eigenvector** of an $n \times n$ matrix A is a nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ . A scalar λ is called an **eigenvalue** of A if there is a nontrivial $\mathbf{x}$ of $A\mathbf{x} = \lambda\mathbf{x}$; such an $\mathbf{x}$ is called an *eigenvector corresponding to λ* .



Example 1

Let $A = \begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix}$, $\mathbf{u} = \begin{pmatrix} 6 \\ -5 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$. Are $\mathbf{u}$ and $\mathbf{v}$ eigenvectors of A ?



Example 2

Let $A = \begin{pmatrix} 3 & 4 \\ 6 & 5 \end{pmatrix}$. Are -1 and 3 eigenvalues of A ?



Definition

The set of all solutions of $(A - \lambda I)\mathbf{x} = \mathbf{0}$ is called the **eigenspace** of A corresponding to λ .



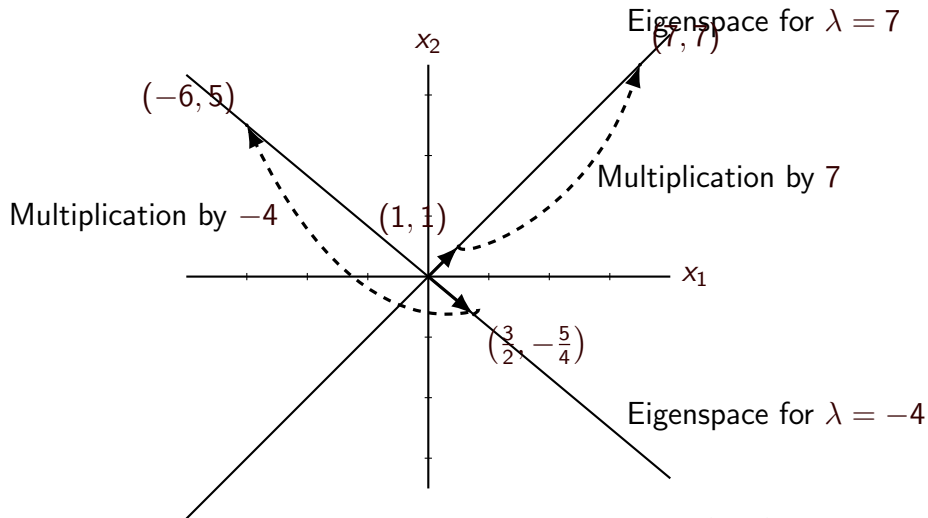


Figure 2. Eigenspaces for $\lambda = -4$ and $\lambda = 7$, with $A = \begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix}$.

Remark

λ is an eigenvalue of $A \Leftrightarrow (A - \lambda I)\mathbf{x} = \mathbf{0}$ has ontrivial solution $\Leftrightarrow \det(A - \lambda I) = 0$.

Definition

- $\det(A - \lambda I) = 0$: a characteristic equation of a matrix A .
- $p_A(\lambda) = \det(A - \lambda I)$: a characteristic polynomial of A .

λ is an eigenvalue $\Leftrightarrow \lambda$ is a solution of a charateristic equation.



Definition

The (algebraic) multiplicity of an eigenvalue is its multiplicity as a root of the characteristic equation.

Definition

The (geometric) multiplicity of an eigenvalue is the dimension of corresponding eigenspace.

Theorem

The algebraic multiplicity of an eigenvalue is greater than or equal to its geometric multiplicity.

Notice The geometric multiplicity is greater than or equal to 1 .



Example

The characteristic polynomial of a 6×6 matrix is $\lambda^6 - 4\lambda^5 - 12\lambda^4$. Find the eigenvalues and their multiplicities.

Solution. Factor the polynomial

$$\lambda^6 - 4\lambda^5 - 12\lambda^4 = \lambda^4(\lambda^2 - 4\lambda - 12) = \lambda^4(\lambda - 6)(\lambda + 2)$$

The eigenvalues are 0 (multiplicity 4), 6 (multiplicity 1), and -2 (multiplicity 1).



Example

Consider $A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$. The eigenvalues are the diagonal entries $\lambda = 2$ and $\lambda = 3$.

- Eigenspace for $\lambda = 2$:

Solve $(A - 2I)\mathbf{x} = \mathbf{0} \Leftrightarrow \mathbf{x} = (s, t, 0)$, $s, t \in \mathbb{R}$. Thus the eigenspace is $E_2 = \text{span}\{(1, 0, 0)^T, (0, 1, 0)^T\}$. Therefore, the geometric multiplicity of eigenvalue 2 is $\dim(E_2) = 2$.

- Eigenspace for $\lambda = 3$:

Similarly, the geometric multiplicity of eigenvalue 3 is $\dim(E_3) = 1$.



Triangular matrices

Theorem

The eigenvalues of a triangular matrix are the entries on its main diagonal.



Characteristic polynomials

- A is a 2×2 matrix:

$$p(\lambda) = \lambda^2 - \text{Trace}(A)\lambda + \det(A).$$

- A is a 3×3 matrix:

$$p(\lambda) = \lambda^3 - \text{Trace}(A)\lambda^2 + (C_{11} + C_{22} + C_{33})\lambda - \det(A),$$

where C_{11}, C_{22}, C_{33} are the $(1, 1), (2, 2), (3, 3)$ -cofactors of A .



Linear Independence

Theorem

If $\mathbf{v}_1, \dots, \mathbf{v}_r$ are eigenvectors that correspond to distinct eigenvalues $\lambda_1, \dots, \lambda_r$ of an $n \times n$ matrix A , then the set $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly independent.



Example

Matrix $A = \begin{pmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{pmatrix}$ has 3 eigenspaces

$E_3 = \langle (1, 0, 0)^T \rangle$, $E_0 = \langle (-2, 1, 0)^T \rangle$, $E_2 = \langle (-10, 3, 1)^T \rangle$. Then 3 vectors $(1, 0, 0)^T$, $(-2, 1, 0)^T$, $(-10, 3, 1)^T$ are linearly independent.



Linear Independence

Theorem

Let A be a square matrix of size n . If A has k different eigenvalues $\lambda_1, \dots, \lambda_k$ with corresponding eigenspaces: $E_{\lambda_1} = \langle S_{\lambda_1} \rangle, \dots, E_{\lambda_k} = \langle S_{\lambda_k} \rangle$, where $S_{\lambda_1} = \{\mathbf{u}_{11}, \dots, \mathbf{u}_{1m_1}\}, \dots, S_{\lambda_k} = \{\mathbf{u}_{k1}, \dots, \mathbf{u}_{km_k}\}$. then $S = \{\mathbf{u}_{11}, \dots, \mathbf{u}_{1m_1}, \dots, \mathbf{u}_{k1}, \dots, \mathbf{u}_{km_k}\}$ is linearly independent.



Example

Matrix $A = \begin{pmatrix} 3 & 6 & -8 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ has 2 eigenspaces

$E_3 = \langle (1, 0, 0)^T \rangle$, $E_2 = \langle (-6, 1, 0)^T, (8, 0, 1)^T \rangle$. Then three vectors $(1, 0, 0)^T, (-6, 1, 0)^T, (8, 0, 1)^T$ are linearly independent.



Theorem

If $n \times n$ matrices A, B are similar, then they have the same characteristic polynomial and hence the same eigenvalues (with the same multiplicities).

$A, B \in M_n(K)$ are called similar if there exists an invertible matrix P , such that $B = P^{-1}AP$.

Remark $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ are not similar even though they have the same eigenvalues.



Some properties

Let λ be an eigenvalue and $\mathbf{x}$ be the corresponding eigenvector of matrix A . Then

- λ is an eigenvalue of A^T .
- $c\lambda$ is an eigenvalue of cA .
- λ^k is an eigenvalue of A^k .
- λ^{-1} is an eigenvalue of A^{-1} . Here $\mathbf{x}$ is also the corresponding eigenvector of these matrices.

